



**VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2015
Higher 2**

Answer Scheme

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CANDIDATE NAME

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BIOLOGY

9648/03

Applications Paper and Planning Question

Paper 3

18 September 2015

Additional Materials: Answer Booklet/ Paper

2 Hours

READ THESE INSTRUCTIONS FIRST

Write your Class, exam number and name on all the work you hand in.
Write in dark blue or blue pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use any staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

Answer **all** questions.

- 1 One of the first potential targets for biotechnology products for animal products was the use of bovine somatotropin (bST) in cows. bST influences the tissue metabolism so that more nutrients can be channelled into milk synthesis. This allows the animals to achieve an increased milk yield while remaining normal and healthy. bST is widely marketed to dairy farmers in the US, who can add the hormone as a form of supplement to the diet of their cows.

A student wanted to carry out an experiment to clone the *bST* gene for expression. **Fig.1.1** below shows a map of the region of the cow's genome where the *bST* gene is located and **Fig. 1.2** shows the plasmid that he intends to use as a vector.

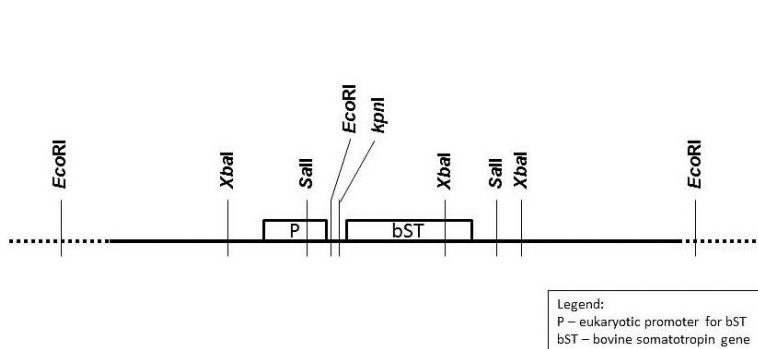


Fig.1.1

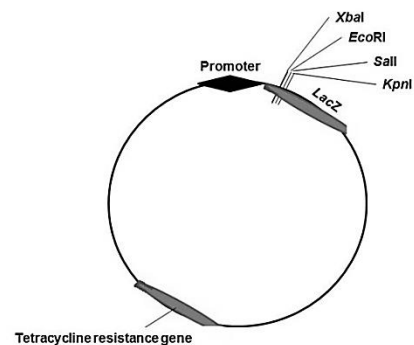


Fig. 1.2

- (a) (i) With reference to **Fig. 1.1** state and explain the most ideal combination of restriction enzymes that you would use to isolate the *bST* gene for expression. [3]
- *EcoRI* and *SalI*; (R! *KpnI* and *SalI* because insertion will be in the wrong direction, or if students use only 1 RE as question asks for a combination of them.)
 - Using both RE allow the **entire *bST* gene** to be **released intact**;
 - While the **different complementary sticky ends** at each end;
 - Position of the restriction sites: *EcoRI* located on the upstream/ nearer to the promoter while the *SalI* located further away from promoter or *EcoRI* located on the upstream of *SalI*;
 - would ensure that the *bST* gene is inserted in the **correct orientation** into the plasmid (where the promoter is upstream of the 5' of the gene);
- (ii) Describe how two features of the plasmid shown in **Fig. 1.2** would allow him to identify the bacteria that have taken up the *bST* gene. [3]
- **Tetracycline resistance gene**; [1m]
 - allows bacteria that have taken up the *bST* gene to **survive** in **medium** containing **tetracycline**;
- (Hence distinguishing between transformed and non-transformed bacteria/ selects for

transformed bacteria with bST.)

- **LacZ gene**; [2m]
- Which codes for **β -galactosidase**; (accept reverse argument when insertionally inactivated)
- Is **insertionally inactivated**;
- Hence cannot break down X-gal into blue product, colonies are **white** in colour;

(Can differentiate between transformed bacteria with annealed plasmid and recombinant plasmid.)

(iii) He cultured the bacteria containing the **bST** gene in an incubator but no hormone was produced. Explain how this problem can be solved. [2]

- The *bST* gene is eukaryotic in nature and **contains introns**, which **cannot be excised** by prokaryotic **bacteria** as they **lack post-transcriptional machinery**;;
- Isolate **mature bST mRNA** and **reverse transcribed** to obtain a double stranded **cDNA** for insertion into vector for bacteria transformation;;

(b) Commercial production of bST makes use of fermenters containing cultures of cells expressing the gene.

Suggest two reasons why this method of producing bST is preferred over insertion of **bST** gene directly into the genome of the cow. [2]

Any two below:

Bacteria

- Easier to control and manipulate the growth conditions of the fermenter to stimulate optimum production;;
- More bST produced per unit time/ faster rate of production as bacteria multiply very fast via binary fission resulting in many copies of the genes for expression;;
- Large scale/ mass production of the bST resulting in lower cost/ more cost effective;;
- Hormone can be produced in large quantities, pure form, at relatively low cost;;
- No ethical concern here of having to genetically modify animals;;

Cow

- Random insertion of the gene into the cow's genome may result in **insertional mutagenesis** when the bST gene disrupts TSG/proto-oncogenes/critical genes;;
- bST gene **may not be expressed** as it is not inserted into the genome/ inserted into non-coding region
- Use of **viral vector/ retroviruses** in the process may lead to insertional mutagenesis or cause an immune response in the cow, leading to decrease or no bST production;;

- Gene may not be inserted into the genome, resulting in transient expression and hence low hormone production;;
- AVP

- (c) With the advancement in genetic engineering techniques, more food has been genetically modified. Although the US Food and Drug Administration have stated that such food is safe for human consumption, some consumers are demanding that such food be labelled as genetically modified (GM) food.

Discuss briefly one ethical concern about not labelling genetically modified food. [2]

- Consumers have the **right** to make **informed decision/ choice** (about the food they are eating);; and

Any one point below:

- Consuming GM food that contain genes that are against their beliefs
 - Some religious groups with strong dietary restrictions - Consumption of food with **genes from animals** that the **religion considers unacceptable;;**
 - Consumers who are against genetically modifying animals and plants;;
- Consuming GM food that contain genes that can cause side effects
 - Food may contain **allergens for some consumers** with **food allergies** e.g. peanut;

- 2 Tissue culture is a technique that can be used to propagate plants that are difficult to grow from seeds. Explants are removed and cultured in nutrient rich medium to produce calluses.

- (a) (i) Define the term *callus* and explain how its features have been exploited in tissue culture. [3]

- Callus is a mass of **undifferentiated** cells;
- Which are **totipotent** and can **give rise to a whole plant**;
- Tissue culture is **in vitro propagation/ cloning** of these totipotent cells;
- To produce huge numbers of **genetically identical plantlets**;
- Through manipulation/ adjustment of culture media that contain **plant growth regulators** such as auxins and cytokinins;
- That will promote **differentiation into specific plant structures**;

(ii) Suggest two advantages of tissue culture over the more traditional methods of cloning plants, such as taking cuttings or grafting. [2]

- TC is **relatively rapid** in producing plant of desired trait compared to grafting;;
- TC makes **mass propagation** of genetically identical plants with desired

trait possible;;

Accept:

- TC takes up less space;;/ use of meristematic regions ensures virus-free plants can be cloned;;/easier to control conditions in the lab vs conditions in the field/ natural conditions;;

Reject:

- TC is performed in sterile conditions so that the plants will be free from pathogens.
- TC allows for formation of genetically identical plantlets since cuttings and grafting from the same plant also produces clones i.e. genetically identical plants as stated in the question.
- TC allows propagation of plant species that are difficult to propagate by cuttings and grafting. The context is that they are established methods for cloning, so TC is advantageous because of speed and quantity rather than difficulty.

(b) Nutrient media formulations for tissue culture must include:

- a source of carbon
- various essential minerals
- vitamins
- growth regulators

The rates of growth of rattan explants in two types of regulators are shown (**Fig. 2.1**).

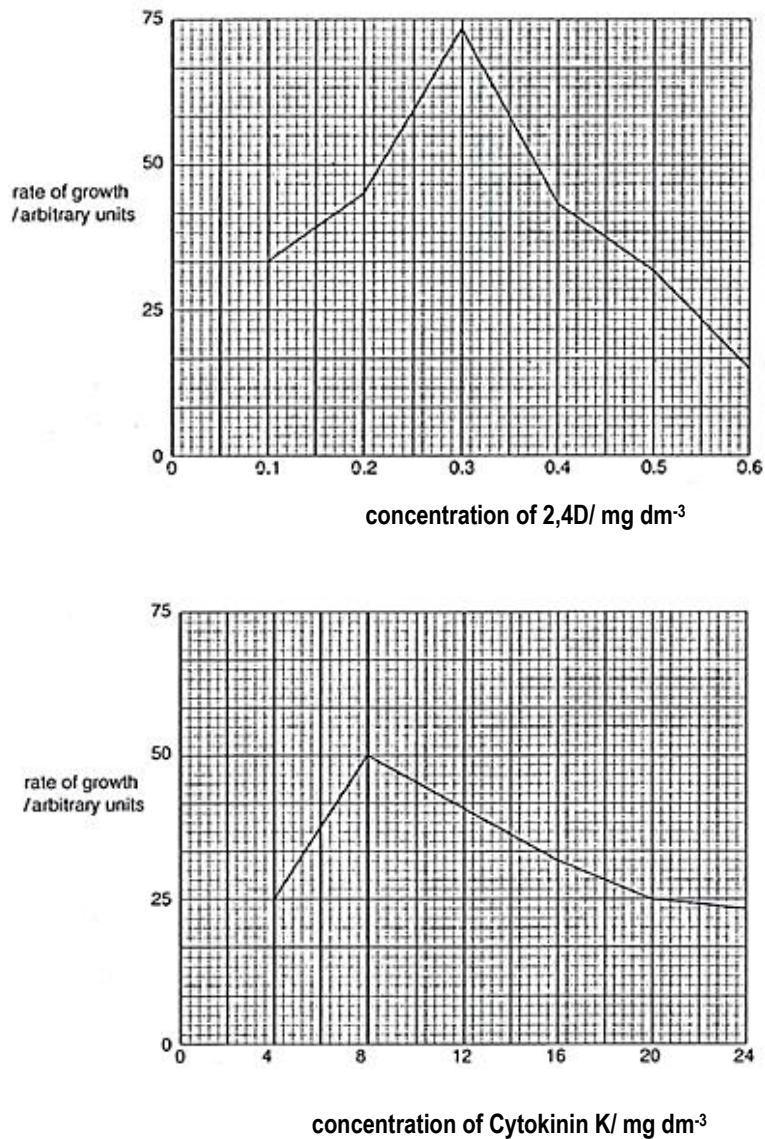


Fig. 2.1
Adapted from N07 A level

- (i) With reference to **Fig. 2.1**, compare the extent to which variations in the concentrations of **2,4 D** (a synthetic auxin) and **cytokinin K** affect the successful growth of the explant. [4]

Similarities (max 2m):

- **Both 2,4D and cytokinin K increase the rate of explant growth up to a certain concentration** (QV – $0.1\text{-}0.3 \text{ mgdm}^{-3}$ and $4\text{-}8 \text{ mgdm}^{-3}$, respectively) **then rate of explant growth decreases;;**
- Increasing the concentrations of **both plant regulators beyond the optimum concentration have a sharply negative effect** on explant growth (QV- above 0.3 mgdm^{-3} for 2,4D and above 8 mgdm^{-3} for cytokinin K);;

Differences (max 2m):

- **Growth rate is more sensitive to changes in 2,4D concentration** compared to changes in cytokine K (QV- increase of 0.2 mg dm^{-3} of 2,4D increases rate from 33 to 75 au compared to increase from 4 mg dm^{-3} of cytokinin K increases rate from 25 to 50 au);;
- **2,4D is more effective** in promoting successful growth of rattan explant as **maximum rate** (QV- 75au) **is higher** than maximum rate for cytokinin (50au) irrespective of concentrations;;
- Growth rate **continues to decrease at high concentration of 2,4D** (QV- $0.5\text{--}0.6 \text{ mg dm}^{-3}$) whereas rate of growth **levels out with cytokinin K** ($20\text{--}24 \text{ mg dm}^{-3}$);;
- AVP;;

(ii) **2,4 D** has been used to control the growth of weeds because it is also able to cause the death of cells. Use the information in **Fig. 2.1**, to justify what might be the lowest effective concentration of **2,4 D** to elicit cell death. [2]

- Just above 0.3 mg dm^{-3} (A! 0.3 mg dm^{-3} if bullet 2 follows);;
- Sharp decrease in growth rate after 75 au (indicates death of callus cells taking place or slower growth and division of cells taking place negative effect of 2,4D (for 0.3 mg dm^{-3});; OR
- 0.45 mg dm^{-3} ;;
- Growth rate is lower than lowest reading (33 au) at beginning of experiment as more cells are dying than dividing;;

(c) Nitrogen is also a vital constituent of tissue culture medium. During the investigation, it was determined that ammonium nitrate is a better source of nitrogen than potassium nitrate for the growth of the explants.

(i) Explain why the explants need a source of nitrogen. [2]

- Nitrogen is an element/ component of **amino acids** which are **building blocks/ monomers of proteins**;;
- Element/ component of **nitrogenous bases** which are **building blocks of nucleic acids such as DNA and RNA**;;

Accept:

- Nitrogen is required for synthesis of chlorophyll.
- Structural components of cells/ enzymes only if it is linked to nitrogen-based biomolecules.

Reject:

- Nitrogen is required for photosynthesis as this is too vague.

(ii) Suggest why ammonium nitrate may be a better source of nitrogen than potassium nitrate. [1]

- **more N** in ammonium nitrate compared to potassium nitrate;;
- N in ammonium nitrate is **more accessible** / **more easily taken up** by cells;;
- AVP;;

[Total: 14]

- 3 Many marine fish species that inhabit icy sea waters synthesize antifreeze proteins (AFP) to protect them from freezing. The *AFP* gene codes for the antifreeze protein in the ocean pout. A transgenic Atlantic salmon line is created by placing the Pacific chinook salmon growth hormone (GH) cDNA driven by the AFP promoter that was taken from the ocean pout. This transgenic salmon is called AquAdvantage salmon.

The ocean pout antifreeze promoter is constitutively (i.e. continually) expressing in all tissues of the transgenic salmon. In contrast, the growth hormone promoter in standard salmon is only active in response to certain environmental cues such as day length and temperature in some tissues.

The growth of AquAdvantage salmon and the wild type (standard) salmon was measured after the first feeding and the results are shown in **Fig. 3.1**.

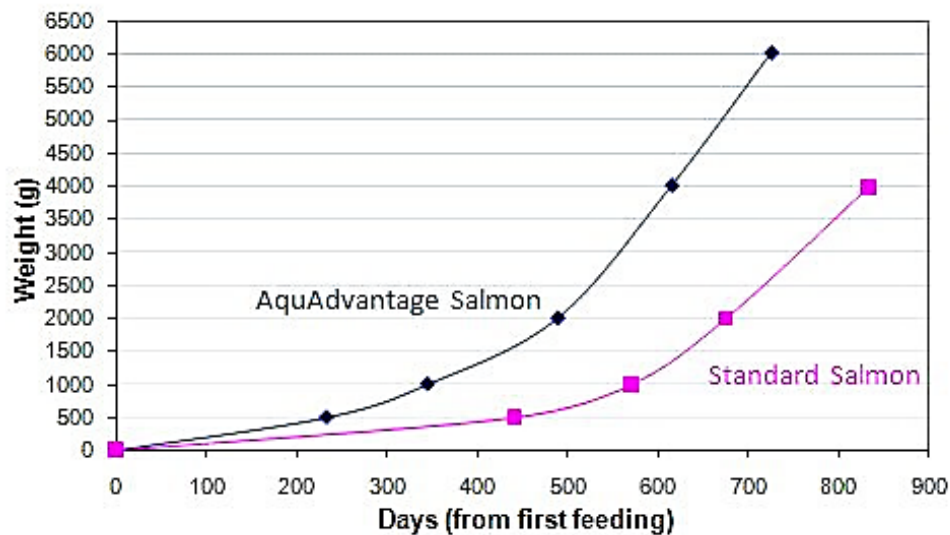


Fig. 3.1

- (a) Use the context of AquAdvantage salmon to explain the term **genetically modified organisms**. [2]

- GMOs – organisms that have **acquired one or more genes** (note - from a different species of fish in the case of AquAdvantage salmon) by **artificial means/ via genetic engineering techniques**;;
- In AquAdvantage salmon, the **foreign DNA** is derived from **non-coding**

sequences of antifreeze gene from the ocean pout and the **coding sequence of growth hormone gene** from the Pacific Chinook salmon;;

Accept:

- “Foreign genetic material” only if gene sequences is mentioned later. GMOs have to express proteins to have a phenotype;
- AFP gene as it is the promoter and 3’UTR sequences;

(b) (i)Based on **Fig. 3.1**, and the information provided, describe the growth of transgenic salmon compared to standard salmon. [2]

- **Both fishes show increasing trend** in weight after feeding;
- **Increase in weight** was **relatively modest initially** at 250 days, transgenic salmon reaches 500 g compared to standard salmon which is slightly lower at 250 g;

Accept any one below:

- Increase **more sharply** from 250 days to 700 days (500g to 6 Kg) while standard salmon increases modestly (250 g to 2.25 Kg) by the same time;;
- **Gap between increase in weigh widens sharply after 500 days** / gradient becomes steeper / By 850 days, standard salmon has not reached 6 Kg;;

(ii)Relate the possible effect of the AFP promoter on growth rate observed in transgenic salmon as compared to standard salmon. [2]

- Sufficient for expression of GH to produce the **rapid growth** phenotype;
- As the **promoter is constitutively/ persistently active** in transgenic salmon **independent of environmental conditions/ A!** at all times;
- Promoter is also **active in almost all tissues** as GH transgene mRNA was expressed in almost all body tissues;
- Transgenic GH hormone effects **rapid growth** in transgenic vs **standard which does not express GH gene at all times**;

The development of the AquAdvantage Salmon began in 1989 with the micro-injection of the transgene into fertilized Atlantic salmon eggs.

Fig. 3.2 shows the structure of the transgene. It contains the coding sequence of the growth hormone gene from the Pacific Chinook salmon under the control of 5’ UTR, promoter and 3’ UTR sequences of the antifreeze *OP5a* gene from the ocean pout.

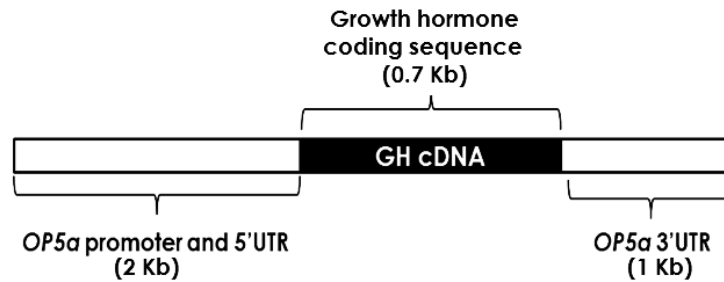


Fig. 3.2

Using the information provided, outline briefly how one would use PCR to distinguish between a transgenic salmon and a standard salmon. [4]

- Synthesise **forward and reverse primers** that are **complementary** to the 3' end of the two strands containing **OP5a 5' and 3' UTR sequences** ;;
- **Isolate DNA from transgenic and standard salmon** and subject to PCR using **primers, deoxyribonucleotides (+ATP)** using **25-30 cycles** of **denaturation, annealing and extension** steps to **amplify target region**;;
- **Gel electrophoresis of PCR product** and then stained with **ethidium bromide** and observe under **uv light**;;
- **Transgenic salmon** will give **one band of expected size (QV – 3.7 Kb)** whereas **standard salmon** will **not give any band** since it **does not have OP5a sequences**;;

R! backward primer

The AquaAdvantage salmon is produced as a triploid and therefore sterile. Transgenic eggs are generated in the AquaBounty Technologies facility on Prince Edward Island (Canada) before shipping to a restricted facility in Central America, where they will be reared. Once they reach market size (**Fig. 3.1**), they will be harvested for processing (e.g. preparation of fish fillets, steaks, etc.) for consumption. No live fish will be sold to consumers.

- (d)** Identify one social and one ethical concern surrounding genetically modified organisms and discuss whether AquaAdvantage salmon successfully addresses these concerns. [4]

Social concern @ 2m

- GMOs pose risks to the safety to humans e.g. **potential sources of allergens**;;
- transgene product GH is derived from a **natural Chinook species from the wild** so is **not likely to be a health risk**;;
- GMOs pose risk to safety of the environment by **upsetting the ecological balance**;;

- Transgenic fish may **escape and establish themselves in the wild** and may have **advantage of large size and rapid growth in competing for natural resources** used by the natural species, thereby **affecting natural species survival**;;
- GMOs pose risk to safety of the environment by **upsetting the ecological balance by interbreeding with WT** Atlantic salmon;;
- Transgenic fish **produced as triploid, thus sterile** ► will not produce viable gametes ► will not expand population through prezygotic barrier ► no fertilisation to produce viable zygotes;;
- GMOs pose risk to safety of the environment by **disrupting natural food chain**;;
- Transgenic fish **are reared in controlled/ restricted culture facility** until market size before harvesting for **processing to fish products** for sale;; / **no retail of live fish**;;

Ethical concern @ 2m

- Exploitation of animals for food is unacceptable/ **tampering with nature** via cross-species gene-mixing by scientists is a concern to many religious groups;;
- AquAdvantage salmon is produced by artificial means so cannot address this concern;;
- Questionable **right to patent GMO** particular sectors of public that take issue with this ethical concern;;
- AquAdvantage salmon is patented for commercial production/ profit so cannot address this concern;;
- **Genetic manipulation of animals (and plants) is not acceptable** amongst sectors of public concerned about possible harmful effects of transgene expression on development of the fish;;
- No information provided on AquAdvantage salmon development and health so cannot address this concern ;;
- AVP – e.g. dietary issues with reasons;; profit motive over-riding other important factors with reasons;;

[Total:13]

4 Planning question

You are required to plan but not carry out, an investigation into the effect of different wavelengths of light on the rate of photosynthesis in two varieties of the same species of a garden plant. These two varieties differ in their ability to produce chlorophyll b.

Variety **A** Normal wild type (presence of chlorophyll b)

Variety **B** Chlorophyll b deficient

Leaf extracts from each variety are prepared by grinding 10 g of the leaf sample in 30 cm³ of ice cold phosphate buffer solution. The suspension is then filtered through a fine mesh and the filtrate covered with aluminum foil and placed in an ice bath.

You are provided with thin capillary tubes that can be used to draw up the filtered leaf extract.

The method used to find out the rate of photosynthesis is to determine the time taken for blue DCPIP added to the filtrate to decolourize. When oxidized, DCPIP is blue in colour and it turns colourless when it accepts electrons and becomes reduced.

Figure 4.1 below shows the setup for the experiment.

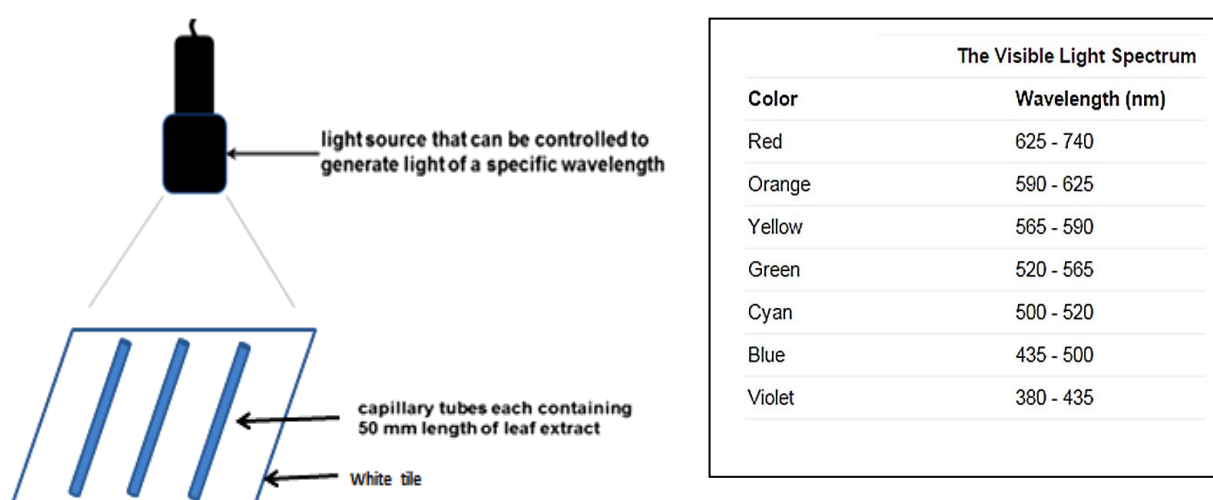


Figure 4.1

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use.

- Filtered leaf extracts from the two varieties in beakers covered with aluminum foil (sitting in ice)
- A bench light source that is fitted with refractors and filters to enable you to get a specific wavelength known as monochromatic light
- 1% DCPIP solution

- Capillary tubes (50 mm in length)
- Timer e.g. stop watch or stop clock
- Petri dishes
- White tile
- Ruler
- Aluminum foil
- Glass rods

You can also select from the following apparatus:

- A variety of different sized beakers, measuring cylinders or syringes for measuring volume
- teat pipettes

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s) if necessary,
- identify the independent and dependent variables,
- include full details and explanations of the procedures that you would adopt to ensure that the results obtain will be as quantitative, precise and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms.

[Total:12]

Planning Question Answers

1. Chlorophyll pigment b **absorb light for excitation of electrons**;
2. absorb mainly **red and blue-violet** light OR 600-700 nm and 400-500 nm;
3. Excited electrons from photosystem can reduce DCPIP to colourless;
4. Greater absorption of light, more excited electrons released from photosystem, shorter time taken for decoloration of DCPIP;
5. Variety A should have a **higher rate of photosynthesis / shorter time required for decolorisation** at the red and blue-violet light; (@ opposite)
6. (a) Dependent variable - **Rate of photosynthesis as measured by time taken** for the decolorisation of DCPIP (1/time); (b) Independent variable - **wavelengths of light**;
7. Diagram showing labelled capillary tubes; fixed distance between light source to capillary tubes;
8. Specify at least 5 different wavelengths of light; (must include red and blue-violet)
9. Factors to be kept constant –
 - (a) Specify volume / no. of drops of DCPIP added;
 - (b) Specify volume / height in tube of leaf extract taken up;
 - (c) Specify distance of light source to tubes (//point 8);
 - (d) Same timing of exposure of extract to light / keep capillary tubes covered with foil;
 - (e) Setting up of a capillary tube without DCPIP added as color standard for comparison;
10. Logical sequence of events –
 - (a) Addition of DCPIP to leaf extract before removing leaf extracts using capillary tubing;
 - (b) Stirring of leaf extract before drawing with the capillary tubing;
 - (c) Start timing when light is on and stop timing when DCPIP is decolourised;
11. Control - [what] capillary tube covered entirely under foil [how] repeat the same procedures as the other tubes [why] to show that decoloration of DCPIP observed in other tubes is due to the absorption of electrons excited by light;;
12. (a) Replicates (replicate tubes must be included in procedures); (b) Repeat entire experiment twice;
13. Table - (a) indep variable in first column; (b) process data for rate (1/time);
14. Graph - (a) correct X and Y axis with units (b) 2 graphs showing variety A and B; (c)

correct trend of graphs;

S. use of key terms such as light reaction centre, photosystem, light dependent reactions, electron transport chain, final electron acceptor, correct dependent independent variable, photolysis of water (any 3);

T. Correct use of all apparatus terminology and functions;

Proposed answer:

In the **light dependent reaction** of photosynthesis, light is absorbed by the **photosynthetic pigments** of the leaves to excite electrons from photosystems I and II. The main photosynthetic pigments involved are chlorophyll a, b which absorbs mainly **red and blue-violet** light/ 600-700 nm and 400-500 nm) and reflects green light/ 500-575 nm.

Electrons released by the activated photosystems can be accepted by DCPIP which becomes reduced and decolourise/ turn colourless. (Thus rate of photosynthesis can be determined by the time taken for DCPIP to decolorise – provided in qn)

The higher the absorption of light, the greater the number of electrons released from the photosystems and hence the shorter the time taken to decolourise the DCPIP.

Variety A should have a **higher rate of photosynthesis** at the red and blue-violet light with lower rate in the green light. Variety B should have an overall lower photosynthetic rate at the same wavelengths due to the lack of chlorophyll b.

Dependent variable: Rate of photosynthesis as measured by time taken for the decolourisation of DCPIP or 1/time taken; Independent variable: wavelengths of light

1. Stir the contents of the boiling tube with a glass rod before pouring a fixed volume (e.g. 5 cm³) of the leaf extract of Variety A into a petri dish.
2. Take a capillary tube and stand one end of the tube into the leaf extract. Some of the liquid will rise up into the capillary tube. Ensure that the extract fill up the entire capillary tube/ 50 mm. Place the capillary tube on a white tile and cover it with aluminium foil.
3. This serves as a colour standard to confirm whether DCPIP has been decolorized in presence of different wavelengths of light in the other tubes.
4. Using a teat pipette, add **2 drops** of 1% solution of DCPIP to the leaf extract in the petri dish and mix well with a glass rod. (Accept more than 2 drops as long as the number is fixed and specified.)

5. Repeat step 2 with a new capillary tube to draw up the leaf extract+DCPIP mixture.
6. Place this capillary tube containing the leaf extract+DCPIP onto the same white tile from step 2.
7. Completely cover the capillary tubes and petri dish with aluminum foil to shield the contents from light.
8. Prepare a second capillary tube in the same way/ following steps 4 – 7. This is the **replicate**.
9. Repeat step 4 – 7 to create a control tube, but do not remove the aluminum foil at all. This is to serve as a control to show that the effect of DCPIP decolorizing is due to the effects of light being used for photosynthesis.
10. Using a ruler, **fix the distance of the light source** at 15cm above the white tile. (Accept alternative distances as long as the number is logical and fixed.)
11. Adjust the light source to emit a specific wavelength of light eg corresponding to violet light. Turn on the light. Remove the aluminium foil covering the capillary tubings.
12. Start the stop watch. Note and record the time taken for the blue colour to disappear.
13. Repeat the experiment from step 5 to 13 for the following wavelengths of light: ranging from blue, cyan, green, yellow, orange and red.
14. Repeat the experiment from steps 1 to 13 at least twice.
15. Repeat the entire experiment from steps 1 to 14 for the leaf extract from Var B.

Table showing the effectiveness of the different wavelengths of light in decolourising DCPIP for Variety A.

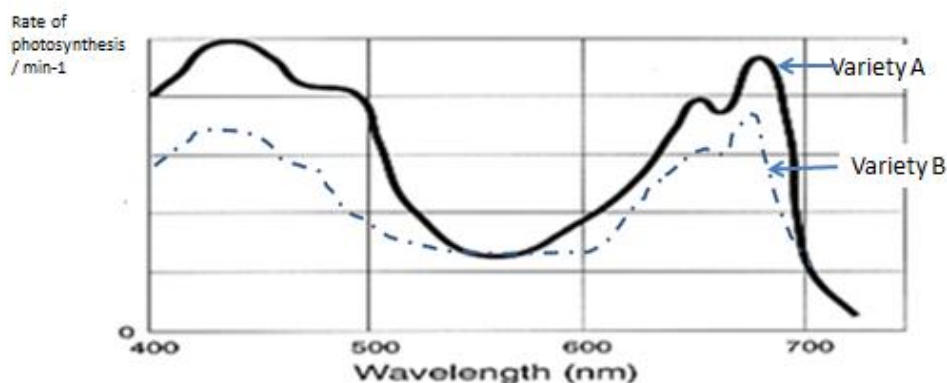
Construct the same table for Variety B

Wavelength of light/nm (Give actual value)	Time taken for DCPIP to decolourised/ min							Rate at which DCPIP decolourised/ min ⁻¹
	Expt 1		Expt 2		Expt 3		Average	
	R1	R2	R1	R2	R1	R2		
Violet								
Blue								
Cyan								
Green								
Yellow								
Orange								
Red								

Average time taken = $[(R1+R2 \text{ of Expt 1}) + (R1+R2 \text{ of Expt 2}) + (R1+R2 \text{ of Expt 3})]/6$

Rate at which DCPIP decolourised = $1/\text{average time taken}$

Graph showing the effectiveness of the different wavelengths of light in decolourising DCPIP.



Feedback:

Theoretical Consideration:

- Many students were unable to accurately address the role of light in photosynthesis in the theoretical considerations. Some tends to over-write by explaining the entire process of light-dependent reactions leading into light-independent reactions.
- Many confused chlorophyll pigment b with the carotenoids and were unable to describe the wavelengths of light it absorb strongly in. Common misconception by students is that chlorophyll a absorbs strongly in red light while chlorophyll b absorbs strongly in blue-violet.
- For description of trends, avoid terms like proportional or inversely proportional as it is not clear. Students should simply state the trend directly. Eg. 'The shorter the time of ____, the faster the rate of ____'

Procedures / Methods:

- The design of the experiment requires students to mix the DCPIP and the leaf extract together first before using the capillary tubes to draw up the mixture. Some designed exp to draw the two solutions separately.
- As a staining reagent / redox indicator, students need to be aware that addition of DCPIP should be minimal (count in drops). There were many who made use of the recommended ratio for enzyme-substrate (E.g. 5:1) which is not appropriate in this case.

- Students need to be aware of the need to specify parameters planned out in the exp. Most students missed out on stating the volume / height of leaf extract + DCPIP mixture in each capillary tube.
- For setting of wavelengths, the values need to take into account of the colour spectrum table provided. For example, suggested wavelengths of 200, 300, 400, 500, 600, 700 would not work in this exp as the colours of light will be restricted to only violet, cyan and red when at least 5 different variables are expected. Note that a range of colours given by students can be accepted.
- There is no need to draw leaf extract + DCPIP mixture out at regular intervals for observation. This is unlike other exp which requires interval testing because (i) there are no tests to be done to the mixture; (ii) any changes to the colour of the mixture can be observed directly through the capillary tube.
- Replicate: Note that the design of a good replicate should be included into the steps of the exp. For eg, preparing two capillary tubes of the same leaf extract + DCPIP mixture under the same wavelength of light rather than merely stating as a last step to 'conduct repeats with replicates of the experiment'.
- Control: A suitable control needs to be able to address the aim of the exp. Since the exp was to investigate the effect of wavelength of light, a more appropriate control would be one without light exposed. Thus controls such as (i) boiled and cooled leaf extract (ii) replace leaf extract with distilled water are all not accepted.
- Safety precaution is not required as question outline does not call for it.
- Graph: Note that the aim of expt is to investigate the effect in two different varieties of plant. Thus there is a need to (i) plot two graphs; (ii) plot both graphs on the same axis for comparison purposes.

5 Free-response question

- (a) Describe how a non-viral gene delivery method can be used to treat Cystic Fibrosis.

[6]

1. Mutation in the **gene coding for Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) protein** causes Cystic Fibrosis which can be treated using **liposome delivery system**;;
2. **DNA-liposome complex** is prepared by complexing/ enclosing DNA containing a copy of wild-type *CFTR* gene with liposome;;
3. The DNA-liposome complex is delivered to the **lung cells** via an **aerosol spray**;;
4. When inhaled by the patient, the DNA-liposome complex **enters lung epithelial cells by endocytosis**;;
5. The liposome is **degraded within the endosome** and the **DNA released into the cytosol**;;
6. The **DNA enters the nucleus** of the cell and **integrates**, by recombination, **into a chromosome** of the target cell (A! replaces the mutant allele);;

- (b) Outline the process of nucleic acid hybridisation and explain how it can be used to detect and analyse restriction fragment length polymorphism (RFLP).

[6]

- **Probes** must be synthesized such that they are **complementary to target sequences of interest**;
- and **labelled with radioactive elements**;
- **Same restriction enzyme**;
- used to treat **polymorphic DNA sequences** to generate **restriction fragment of different lengths**;;
- Restriction fragments **separated by gel electrophoresis**;
- then gel is treated in mixture of **alkali and salt to denature the DNA fragments**;
- Transferred over to **nitrocellulose membrane** via capillary action;
- And **bake it at 80°C** to bound the DNA to the filter
- **Hybridize to single stranded probe**;
- **via complementary base pairing** – ref formation of **H bonds**;
- **Wash off excess probe**;
- **Photographic film or an X ray film** is laid over the filter;

- viewed **with autoradiography**;

(c) Discuss the goals and benefits of the Human Genome Project. [8]

Goals (3m max)

- Identify all the 20,000-25,000 genes in human DNA;;
- Construct a detailed physical map of the entire genome to elucidate / understand the genetic basis of diseases;;
- Determine the sequence of the 3 billion base pairs that make up human DNA for storing in databases accessible to the public;;
- Improve tools for data analysis;;
- Address the ethical, legal and social issues (ELSI) that may arise from the project;;

Benefits (5 max, must include explanation to get full mark with maximum 1m for each category)

- Molecular medicine – for improved diagnosis of disease/ earlier detection of genetic predisposition to disease;;/ drug design based on molecular information;;/ design of GT to cure genetic diseases;;
- Energy sources and environmental applications (microbial genomics) – detection of disease-causing microbes;;/ development of new energy sources such as biofuels;;/ development of techniques to monitor environmental pollution;;/ efficient environmental remediation;;
- Risk assessment – to assess health damage and risks caused by radiation exposure;;/ caused by exposure to mutagenic chemicals and cancer-causing toxins;; /to reduce likelihood of heritable mutations;;
- Bioarcheology, anthropology, evolution and human migrations – study evolution through germline mutations in lineages;;/ study population migration through matrilineal line based on female genetic inheritance or patrilineal line through Y chromosome mutations;;/
- DNA forensics – refer to notes (paternity tests, criminal investigations, exonerate the falsely accused)
- Agriculture, livestock breeding and bioprocessing – refer to examples of GMOs.

[Total: 20]